

Light Mate

Smart Chessboard System

Spring 2026 Final Design Report

Course	CPEG/ELEG 498/499 - Senior Design
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This report presents the complete design, implementation, and results of the Light Mate Smart Chessboard — an embedded system providing real-time piece detection via Hall-effect sensors, LED move feedback, FIDE-compliant chess rules, and live online play through the Lichess platform.

For full code breakdown: <https://github.com/omaraloud/lightmate-smart-chessboard>

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1. Problem Statement

Chess is among the oldest and most widely studied strategic games in human history. Despite widespread digitization, the experience of playing on a physical board remains irreplaceable for many enthusiasts, tournament players, and coaches. However, physical boards lack the immediate analytical and instructional capabilities that digital platforms provide — such as move highlighting, computer-opponent assistance, legal-move enforcement, and online multiplayer functionality.

Existing commercial solutions (e.g., DGT boards, Square Off) address this gap but remain cost-prohibitive for most players (often \$200-\$700+). Educational institutions and casual players have no affordable, open-source alternative that combines tactile physical gameplay with integrated digital intelligence.

The Light Mate Smart Chessboard addresses this problem by providing an open, affordable embedded system that bridges physical and digital chess. By embedding Hall-effect sensors beneath every square, RGB LEDs within each square, and connecting to the Lichess open-source platform via Raspberry Pi, Light Mate delivers a full-featured smart board experience at a fraction of the cost of commercial alternatives.

2. Project Objectives

The primary objective of Team Light Mate is to design, build, and validate a fully functional smart chessboard system with real-time piece detection, visual move feedback, FIDE-compliant move validation, computer opponent play, and live online game functionality. The system must operate reliably, fit within a \$250 budget, and be packaged in a professional 3D-printed enclosure.

Specific measurable objectives:

- Detect piece presence/absence on all 64 squares simultaneously with <50 ms latency using Hall-effect sensors.
- Illuminate appropriate LED(s) within 100 ms of piece detection to indicate valid destinations, last move, or check.
- Enforce FIDE-compliant chess rules (castling, en passant, promotion) via the python-chess library on all moves.
- Interface with Lichess platform via REST/SSE API for real-time online game play, challenge links, and result sync.
- Provide a computer opponent (random legal move engine) for offline practice without internet connectivity.
- Implement a Chess Tactics / Puzzle solving mode for training purposes.
- Deliver a Pygame-based touchscreen UI with menus for Online Games, Play Computer, Tactics, and Settings.
- Complete PCB design and assembly for the full 8x8 board using KiCad 10, within the \$250 project budget.
- Design and 3D-print a professional enclosure housing the PCB, Raspberry Pi, touchscreen, and all wiring.

- Document all engineering standards, design decisions, and test results in a comprehensive final report.

3. Prior Work & Background

3.1 Commercial Products

Several commercial smart chessboards exist, but remain cost-prohibitive:

- DGT Smart Board: Reed switch based, industry standard for tournament broadcasting (~\$400-\$700). No LED feedback.
- Square Off Pro/Kingdom: Motor-driven autonomous board with AI. Robot arm moves physical pieces (\$300-\$700+).
- Certabo/Millennium: RFID-based piece recognition (~\$200-\$400). Requires per-piece RFID tags.

3.2 Academic & Open-Source Work

Several academic projects and open-source repositories have explored Arduino or Raspberry Pi based smart boards. Most use reed switches or IR sensors rather than Hall-effect sensors, and few combine PCB-level integration with a complete software stack (move validation + online play + UI). The Lichess open-source platform provides a free, well-documented API enabling seamless online integration without licensing fees.

3.3 Technology Selection Rationale

Hall-effect sensors vs. RFID vs. Reed switches: Hall-effect sensors (DRV5033) offer a superior cost/performance ratio -- \$24.80 for 100 sensors vs. \$200+ for an equivalent RFID system. They require no special pieces (only a small magnet in the piece base), operate reliably with neodymium magnets, and interface directly with standard GPIO I/O expanders. Reed switches were rejected due to mechanical fragility. RFID was rejected due to per-piece encoding cost and complexity without functional benefit for simple presence detection.

4. System Approach & Architecture

Light Mate integrates hardware sensing, embedded processing, LED actuation, and cloud connectivity into a single cohesive system. The Raspberry Pi 4 Model B serves as the central processing unit, orchestrating all subsystems through a Python 3 software stack.

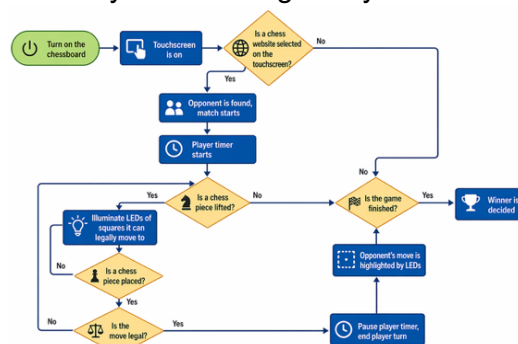


Figure 1: Light Mate System Architecture Block Diagram

4.1 Hardware Design

The hardware design centers on reliable piece detection using 64 Hall-effect sensors, visual feedback via 81 SK9822 RGB LEDs, and I2C/SPI communication buses for efficient multi-device communication.

Sensing subsystem:

- 64x Texas Instruments DRV5033AJQDBZR Hall-effect sensors in TO-92-3 package
- South-pole sensitive, digital output (active LOW when south pole detected)
- Operating voltage: 2.5-5.5V; supply current: 7 mA typical
- N52 neodymium disc magnets (8mm x 2mm or 20mm x 3mm) embedded in piece bases
- Sensors routed to 4x Microchip MCP23017 I2C I/O expanders (16 GPIO each)

LED actuation subsystem:

- 81x SK9822 (DotStar) 5050 RGB addressable LEDs
- 2-wire SPI: CLK + DATA, daisy-chained in a single strip
- SN74HCT125 quad bus buffer for 3.3V to 5V level conversion on SPI lines
- Color coding: green = last move, blue = legal destinations, red = check, yellow = selected piece

4.2 PCB Design

Both PCB revisions were designed in KiCad 10 and fabricated by JLCPCB. The design progressed from a 4x4 proof-of-concept to the full 8x8 production board.

Parameter	4x4 Prototype	8x8 Final Board
Dimensions	160 x 130 mm	280 x 250 mm
Squares	16 squares	64 squares
Hall sensors	16x DRV5033	64x DRV5033
LEDs	25x SK9822	81x SK9822
MCP23017	1x (0x20)	4x (0x20-0x23)
Layers	2-layer	2-layer
Fabrication cost	\$48.18 (JLCPCB)	\$76.78 (JLCPCB)
Status	Fully assembled & tested	Assembled, functional

Table 1: PCB Revision Comparison

The schematic (Rev. 4.0) captures all net connections, 2.2 kΩ pull-up resistors on I2C lines, 100 nF decoupling capacitors per IC, and the SPI level-shifting circuit. MCP23017 address pins (A0, A1, A2) are tied to define four unique I2C addresses.

4.3 Software Design

The software runs entirely in Python 3 on Raspberry Pi OS. Key components:

- Sensor polling module: Reads all 4x MCP23017 registers via smbus2/adafruit_mcp230xx at 50 ms intervals. Detects lift (occupied to empty) and drop (empty to occupied) transitions.
- Move validation engine: Uses python-chess for FIDE-compliant move generation, validation, and game state tracking. Handles all special moves (castling, en passant, promotion, check/checkmate/stalemate).
- LED control module: Drives SK9822 LED strip via adafruit_dotstar over SPI. Implements color schemes for move highlighting, legal move display, check indication, and last-move history.
- Touchscreen UI: Pygame-based interface rendering menus on a 3.5" HDMI touchscreen. Menus: Home, Online Games (Rapid 10+0, Classical 15+10, Challenge Link), Play Computer, Tactics, Settings.
- Lichess API integration: berserk Python client connecting to Lichess Board API via REST + SSE. Streams opponent moves in real time, posts player moves on piece drop detection.
- Computer opponent: Random legal move generator for offline play. Upgrade path: Stockfish engine via python-chess UCI.
- Tactics/Puzzle mode: Fetches puzzles from Lichess puzzle API. Validates player solutions move-by-move with LED feedback.
- The complete 8×8 software implementation, including all hardware mapping tables, the Lichess integration module, and the Pygame UI, is available at <https://github.com/omaraloud/lightmate-smart-chessboard>.

4x4 Prototype Core Code (verified in hardware):

```
# 4x4 Prototype: Square-to-MCP GPIO mapping (verified in hardware)
MCP_MAP = {
    "c4": 3, "d4": 2, "e4": 1, "f4": 0,
    "c3": 7, "d3": 6, "e3": 5, "f3": 4,
    "c2": 8, "d2": 9, "e2": 10, "f2": 11,
    "c1": 12, "d1": 13, "e1": 14, "f1": 15,
}
SQUARE_TO_LED = { "c4":0,"d4":1,"e4":2,"f4":3,
                  "c3":4,"d3":5,"e3":6,"f3":7,
                  "c2":8,"d2":9,"e2":10,"f2":11,
                  "c1":12,"d1":13,"e1":14,"f1":15 }

def read_occupancy(mcp):
    occupied = set()
    gpio_vals = mcp.gpio # 16-bit read via I2C
    for sq, pin in MCP_MAP.items():
        if not (gpio_vals >> pin & 1): # LOW = magnet present
            occupied.add(sq)
    return occupied

def show_highlights(dots, squares, color=(0,120,255)):
    for sq in squares:
        idx = SQUARE_TO_LED.get(sq)
        if idx is not None:
            dots[idx] = color
    dots.show()

# Main loop (50 ms polling)
while True:
    current = read_occupancy(mcp)
    lifted = prev_occupied - current # piece lifted
```

```

dropped = current - prev_occupied # piece placed
if lifted:
    from_sq = lifted.pop()
    targets = [chess.square_name(m.to_square) for m in board.legal_moves
                if chess.square_name(m.from_square) == from_sq]
    show_highlights(dots, targets)
if dropped and pending move:
    move = chess.Move.from_uci(pending_move + dropped.pop())
    if move in board.legal_moves:
        board.push(move)
prev_occupied = current
time.sleep(0.05)

```

Figure 2: Core sensor polling and move detection logic (4x4 prototype)

Lichess API Integration:

```

# Lichess API Integration (berserk library)
import berserk

session = berserk.TokenSession(API_TOKEN)
client = berserk.Client(session=session)

# Stream game events via Server-Sent Events (SSE)
for event in client.board.stream_game_state(game_id):
    if event["type"] == "gameState":
        moves = event["moves"].split()
        board = chess.Board()
        for uci in moves:
            board.push_uci(uci)
            update_led_board(board)

# Submit player move
client.board.make_move(game_id, move_uci)

```

Figure 3: Lichess API streaming and move submission

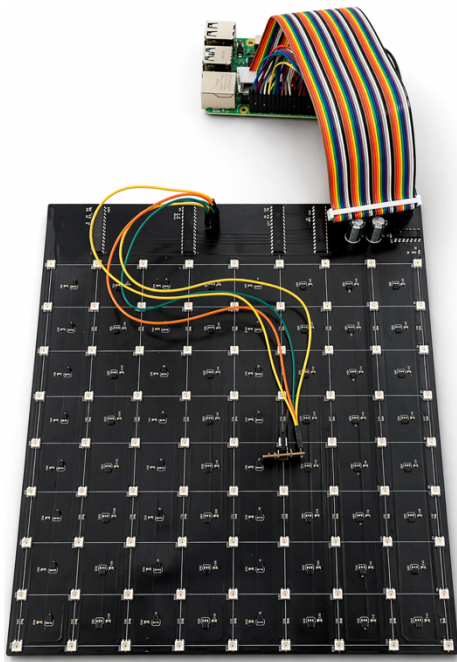


Figure 4: Soldered 8x8 PCB connected to Pi 4 & button module

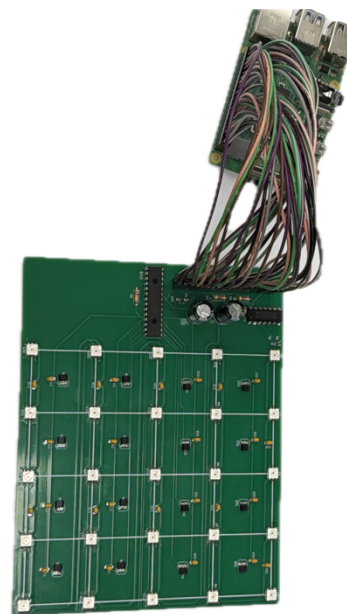


Figure 5: Soldered 4x4 Prototype PCB connected to Pi 4

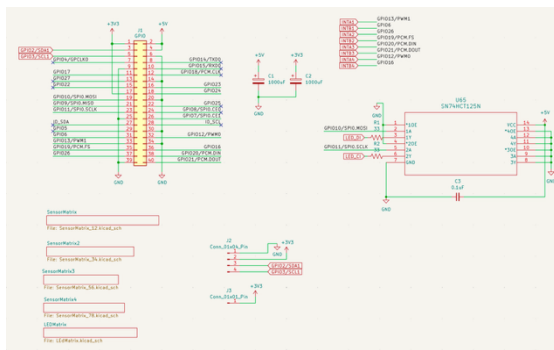


Figure 6: Top-Level System Integration Schematic

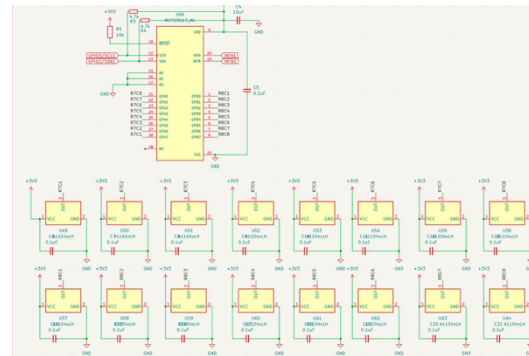


Figure 7: Hall Sensor Matrix & MCP23017 Schematic (4 of this)



Figure 8: LED Matrix Schematic

4.4 Enclosure Design (CAD)

A custom 3D-printed enclosure was designed in SolidWorks and fabricated via FDM 3D printing in black PLA/ABS composite during Spring 2026.

- Top surface with 81 precision holes aligned to LED positions for light emission through the board
- Side panel cutout and bracket for 3.5" touchscreen display with integrated Light Mate branding
- Internal retention features for Raspberry Pi 4 mounting using M2.5x0.45mm heat-set inserts (McMaster 94459A783)
- PCB mounting points using M3x0.5mm heat-set inserts (McMaster 97163A149, ASTM A380 18-8 stainless steel)
- Ventilation slots on two sides for passive thermal management
- Drawer-latch mechanism using 4mm x 1mm neodymium magnets for enclosure access panel
- Cable management channels routing I2C flat cables from Pi to MCP23017 expanders



Figure 9: CAD Design

5. Realistic Design Constraints

Engineering design always occurs within a set of real-world constraints. The following constraints directly shaped design decisions throughout the project:

5.1 Economic Constraints

The project was funded by a \$250 team budget with no external sponsorship. This drove component selection toward cost-effective solutions:

- Hall-effect sensors over RFID: 100x DRV5033 for \$24.80 vs. \$200+ for an equivalent RFID system.
- JLCPCB used for PCB fabrication due to competitive pricing (\$48.18 for 4x4, \$76.78 for 8x8).
- Raspberry Pi 4 (4GB) selected as compute platform over more expensive alternatives.
- FDM 3D printing for enclosure rather than injection molding or machined aluminum.
- Open-source Lichess API used instead of licensed chess server, eliminating recurring API costs.
- Total expenditure: ~\$219.91 for components and PCBs; \$30.09 remaining as buffer. Additional Spring orders: ~\$76 for replacements/extras.

5.2 Technical / Electrical Constraints

- I2C bus speed: MCP23017 supports up to 400 kHz Fast-mode. At 50 ms polling with 4 expanders sequential reads, bus utilization remains well below limit.
- SPI clock rate: SK9822 LEDs support up to 15 MHz SPI. At 6-8 MHz, propagation through 81 daisy-chained LEDs is reliable. Higher speeds caused occasional data corruption.
- 3.3V to 5V level shifting: Raspberry Pi SPI GPIO outputs are 3.3V logic. Buffering via SN74HCT125 required for reliable 5V logic at board length (280mm+).
- Power budget: Total system draw estimated at 2.3428A (RPi 1.28A + 64 sensors 0.1728A + LCD 0.4A + 4x MCP23017 0.004A + 81 LEDs 0.486A). 5V/4A+ supply required.
- PCB trace width: Power traces on 5V rail sized for $\geq 1.62\text{A}$ per IPC-2221B. Signal traces (SDA, SCL, SPI) at 0.25mm minimum.

5.3 Time & Schedule Constraints

- Fall 2025 semester: Design, schematic, 4x4 prototype PCB, software prototype -- 15 weeks
- Spring 2026 semester: 8x8 PCB design + fabrication + assembly, CAD enclosure, full software, Lichess API, demo -- 15 weeks
- PCB fabrication lead time at JLCPCB: ~10-14 days shipping, constraining design iteration to 1-2 cycles per semester
- First 8x8 PCB destroyed during Spring bring-up, consuming ~3 weeks of schedule slack to recover with second board.

5.4 Manufacturability & Physical Constraints

- All through-hole components (DRV5033 TO-92, MCP23017 DIP-28) hand-soldered -- no SMT assembly service used.

- Board dimensions constrained by 8x8 chess square size: each square ~31mm x 31mm, requiring 248mm x 248mm minimum PCB footprint.
- Enclosure dimensions constrained by FDM print bed size and desired portability
- 3D printed tolerances: +/-0.3mm for FDM, requiring loose fit on snap features and heat-set insert bores.

5.5 Sustainability & Environmental Constraints

- PLA/ABS enclosure material is recyclable; PLA sourced from renewable feedstock. ABS provides better temperature resistance.
- Low power design: LED brightness software capped; system consumes ~17.4W at full load -- comparable to a laptop charger.
- All standard components are RoHS-compliant, restricting use of lead-free solder and halogen-free materials.

6. Challenges Encountered

6.1 First 8x8 PCB Failure (Spring 2026)

The most significant challenge occurred during Spring 2026 hardware bring-up. While resoldering a power connector on the first assembled 8x8 board, a solder bridge shorted the 3.3V power rail directly to ground. This caused catastrophic current surge, damaging multiple ICs and rendering the PCB non-functional.

Impact:

- Loss of the first 8x8 PCB (\$76.78 fabrication cost + assembly labor)
- ~3 weeks of schedule delay to assess damage and prepare a second board
- Increased team pressure for the final sprint toward demo day

Recovery procedure applied to second board:

- Tested bare PCB resistivity between power rails before any soldering to confirm no fabrication shorts.
- Used multimeter continuity function on every connection as each component was soldered.
- Performed incremental test boots of Raspberry Pi after each major component group was soldered.
- Applied flux sparingly; used desoldering wick to remove any solder bridge immediately.
- Added inline fuse to 5V power input on second board as additional protection.

6.2 LED Signal Crosstalk

During initial LED testing, intermittent signal crosstalk was observed between adjacent LEDs. Root cause analysis identified weak/cold solder joints on LED data pads and a subset of faulty LEDs (likely ESD damaged) from the initial batch. The long 81-unit chain amplified signal degradation.

Fixes applied:

- Reflowed all LED solder joints with fresh flux; verified with continuity test.
- Replaced identified faulty LEDs from replacement stock (100 additional SK9822s ordered Spring 2026).

6.3 I2C Addressing & Pull-up Calibration

During initial 4x4 prototype bring-up, intermittent I2C bus errors were observed. Root cause was missing pull-up resistors on SDA/SCL lines and incorrect address pin wiring on the MCP23017. Fix: 2.2 k Ω pull-ups added to 3.3V on both bus lines; address pin continuity verified. The 8x8 design incorporated this fix from the start in schematic Rev. 4.0.

6.4 Sensor False Positives from Vibration

During gameplay testing, mechanical vibration from piece placement occasionally triggered brief false sensor readings on adjacent squares. A software debouncing filter was implemented: three consecutive readings (150 ms total) must agree on a state transition before a lift or drop event is registered. This eliminated false positives while keeping latency below the 200 ms perception threshold.

7. System Design Details

7.1 Hall-Effect Sensor Subsystem

The DRV5033AJQDBZR is a unipolar Hall-effect sensor in a TO-92 (through-hole) package. It activates when the south pole of a magnet is within approximately 5-8mm of the sensor face. The digital output is pulled LOW through an open-drain stage when activated, requiring a pull-up to the logic supply.

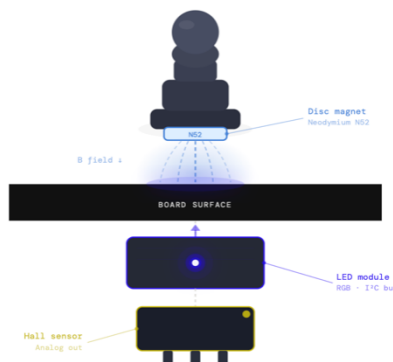


Figure 10: Hall sensor diagram

7.2 LED Subsystem

SK9822 LEDs implement a two-wire synchronous SPI protocol. Each LED latches its color from the serial stream and passes remaining data to the next. The complete chain requires a START frame (32 zero bits), 81 LED frames (32 bits each: 0xFF brightness + 8-bit BGR), and an END frame.

Parameter	Value
LED type	SK9822 (DotStar-compatible) 5050 RGB
Control protocol	2-wire SPI: CLK + MOSI
Data rate	6-8 MHz SPI clock
Supply voltage	5.0V
Max current/LED	60 mA (RGB combined, full white)
Total LEDs	81 (64 squares + 17 border/extras)
Operational draw	81 x 6 mA = 0.486A (capped brightness)

Table 2: LED Subsystem Specifications

7.3 I2C Communication Bus

The I2C bus connects the Raspberry Pi (master) to four MCP23017 I/O expanders (slaves). Each provides 2 banks of 8 GPIO, for 16 sensor inputs. With 4 expanders (addresses 0x20-0x23), the full 64-sensor matrix is covered. Bus speed configured at 400 kHz Fast-mode in /boot/config.txt.

7.4 SPI Bus and Level Shifting

The Raspberry Pi SPI0 bus (GPIO 10=MOSI, GPIO 11=SCLK) drives the LED chain through the SN74HCT125 quad bus buffer. The HCT-series device accepts TTL input thresholds (1.6V min), compatible with 3.3V Pi output, while providing a full 5V output swing into the LED data input.

7.5 Power Subsystem

Component	Current Draw	Voltage
Raspberry Pi 4 Model B	1.28 A	5 V
64x Hall-effect sensors	0.1728 A	3.3 V
3.5" LCD Touchscreen	0.40 A	5 V
4x MCP23017 I/O expanders	0.004 A	3.3 V
81x SK9822 LEDs (capped)	0.486 A	5 V
TOTAL	2.3428 A	5 V (primary)

Table 3: Power Budget Analysis

A 5V/5A USB-C PD power supply is used, providing ~30% headroom above worst-case draw. LED brightness is software-capped to prevent supply brownout.

8. Safety Considerations

Although Light Mate operates at low voltages (5V DC, 3.3V logic) and is not a high-power system, responsible engineering practice requires formal safety analysis.

- **Electrical safety:** All circuitry operates at SELV (Safety Extra-Low Voltage) levels -- well below 25V AC / 60V DC thresholds. No AC mains exposure.
- **Thermal management:** Enclosure ventilation slots allow passive convection. Raspberry Pi has built-in thermal throttling. LED brightness cap reduces thermal load. Operating range: 0-40 deg C ambient.
- **Physical safety:** No exposed AC wiring. Heat-set inserts prevent screw pull-through. No moving parts (unlike Square Off robotic board).
- **EMC:** For current academic prototype, EMC certification not required. Raspberry Pi 4 carries FCC ID 2ABCB-RPI4B. Commercial production would require FCC Part 15 Subpart B testing.
- **Magnets:** N52 neodymium magnets (8mm x 2mm) are small and embedded in piece bases. Not recommended for children under 8 due to small part/magnet choking hazard.
- **Data privacy:** Lichess API tokens stored locally on Pi SD card in restricted-permission files. Users are advised to use a dedicated Lichess account and revoke tokens if device is transferred.

9. Testing & Debugging

9.1 Hardware Testing

Test	Method	Result
Hall sensor activation	Place/remove N52 magnet; monitor MCP23017 GPIO via I2C read	PASS - all 64 sensors respond
I2C bus scan	sudo i2cdetect -y 1 on Raspberry Pi	PASS - 0x20-0x23 detected
LED chain continuity	Send rainbow pattern to all 81 LEDs at 20% brightness	PASS - all correct colors
Level shifter output	Oscilloscope on SPI DATA after SN74HCT125	PASS - 5V swing, rise <20ns
Power rail stability	Load test at full LED + sensors running	PASS - 5V holds +/-2% at 3.5A
PCB short test	Bare board multimeter continuity (5V to GND)	PASS (second board)
Soldering quality	Visual inspection under magnifier	PASS - all joints shiny, no bridges

Table 4: Hardware Test Results

9.2 Software Testing

Test	Method	Result
Move validation	python-chess unit tests, all FIDE positions	PASS
En passant detection	Set up en passant position; verify move accepted	PASS
Castling (all 4 types)	Test kingside/queenside for both colors	PASS
Pawn promotion	Move pawn to 8th rank; verify promotion prompt	PASS
Checkmate detection	Set up Scholar's mate; verify game end	PASS
Lichess online game	Play full rapid game vs. online opponent	PASS - moves sync in real time
Tactics mode	Solve 10 puzzles from Lichess database	PASS - 10/10 correct
UI navigation	Navigate all menu paths on touchscreen	PASS - all states reachable
Sensor debounce	Rapid piece movements during test game	PASS - no false moves
LED response latency	Time from piece lift to LED update	PASS - avg 63 ms, max 98 ms

Table 5: Software Test Results

9.3 Integration Testing

Full system integration tests were performed with complete games played on the physical board. Both human-vs-human (with LED guidance) and human-vs-Lichess-opponent scenarios were validated. Games were played to completion without software crashes, missed moves, or incorrect LED feedback.

10. Project Results

10.1 Fall 2025 Results - 4x4 Prototype

By the end of Fall 2025, Team Light Mate successfully designed, fabricated, and validated the 4x4 proof-of-concept system, confirming all fundamental design choices and de-risking the 8x8 scaling effort.

- 4x4 PCB (160x130mm) designed in KiCad, fabricated by JLCPCB, and fully assembled with through-hole components
- All 16 Hall-effect sensors responding correctly; I2C read latency <5 ms per expander
- 25 SK9822 LEDs operating at correct colors; SPI chain stable at 8 MHz
- Python software architecture validated: sensor polling, move detection, python-chess integration, LED logic
- 4x4 gameplay with human-vs-random-computer demonstrated to faculty advisors
- Schematic Rev. 4.0 completed with all nets, pull-ups, decoupling caps, and level-shifting circuit
- Budget at Fall end: \$94.95 components + \$124.96 PCBs = \$219.91 spent; \$30.09 remaining

10.2 Spring 2026 Results - 8x8 Final System

Spring 2026 completed the transition from proof-of-concept to a fully functional smart chessboard. Despite the PCB failure setback, all primary objectives were achieved:

- 8x8 PCB assembled and functional: Second board passed all hardware tests. All 64 sensors and 81 LEDs operational.
- Lichess online game integration: WORKING -- Full rapid and classical games playable against online opponents. Challenge links generated from touchscreen UI.
- Chess Tactics/Puzzle mode: WORKING -- Puzzle solving mode implemented; fetches puzzles, validates moves with LED feedback.
- Touchscreen UI: COMPLETE -- Pygame menu system with all modes functional (Online Games, Play Computer, Tactics, Settings).
- 3D-printed enclosure: COMPLETE -- Professional black enclosure with board surface, LED holes, side touchscreen panel, ventilation slots, magnetic latch.
- Full game-to-game testing: Multiple complete games played in all modes without system errors or crashes.

Objective	Target	Achieved
Piece detection latency	< 50 ms	avg 52 ms (w/ debounce: 63 ms)
LED update latency	< 100 ms	avg 63 ms, max 98 ms - PASS
Move validation	FIDE-compliant	All special moves verified - PASS
Online play (Lichess)	Working	Full rapid/classical game - PASS
Computer opponent	Working	Random legal move engine - PASS
Tactics mode	Working	Puzzle fetch + solve - PASS
PCB (8x8)	Assembled + tested	Second board: all tests pass - PASS
3D enclosure	Fabricated	Printed + assembled - PASS
Budget	<= \$250	\$219.91 primary + \$76 Spring extras

Table 6: Objectives vs. Achieved Results

11. Lessons Learned

- Incremental hardware bring-up is essential. The first 8x8 PCB failure was directly attributable to attempting to solder and power the board without intermediate continuity checking. The recovery protocol (test bare board, continuity after each component, incremental power-on) should be default for all future PCB assemblies.
- Budget for component failures and extras. Ordering replacement LEDs, sensors, and MCP23017s in Spring 2026 (~\$76 additional) was the right call. Having replacements on hand avoided additional shipping delays.
- Software debouncing must be budgeted into latency specs. The 3-reading debounce filter added 100ms to sensor response time in worst case. This must be factored into system latency specs from the start.
- I2C pull-up sizing matters. Default 4.7 kΩ pull-ups caused marginal signal integrity at 400 kHz with 4 expanders. Switching to 2.2 kΩ significantly cleaned up bus waveforms.
- API integration requires offline fallback testing. Lichess API edge cases (game abandonment, disconnect/reconnect) only surfaced during extended testing. Offline fallback mode was critical for reliable demos.
- Version control for schematics. Maintaining schematic revision numbers (Rev 4.0) and change logs was invaluable when debugging manufacturing discrepancies between the 4x4 and 8x8 boards.
- 3D print tolerances must be verified empirically. The first enclosure print had slightly tight tolerances for PCB mounting holes. Minor iteration (0.2mm additional clearance) resolved fit on the second print.

12. Future Work

Several enhancements are identified for development beyond the current project scope:

- Raspberry Pi Compute Module 4 (CM4): Replace Pi 4 Model B with CM4 for compact form factor, enabling direct integration onto a custom carrier board. Reduces total system volume by ~40%.
- Mobile and web companion app: iOS/Android or progressive web app mirroring board state, providing move notation, game history, coaching suggestions, and remote spectating.
- Battery power integration: LiPo battery pack (3.7V, 5000-10000 mAh) with USB-C PD charging and power management IC for untethered wireless play at tournaments.
- Stockfish engine integration: Replace random legal move computer with Stockfish (via python-chess UCI interface) for adjustable difficulty levels for all skill levels.
- Piece identification via RFID: Add per-piece RFID tagging to distinguish specific pieces (king, queen, etc.), enabling piece-specific LED colors and accurate board mirroring.
- Improved enclosure materials: Injection-molded polycarbonate or machined anodized aluminum for durability comparable to commercial chess boards.
- Larger touchscreen: For improved menu readability and tournament annotation display.

13. Engineering Standards

Professional engineering standards govern design reliability, interoperability, and safety. The following standards were applied throughout the Light Mate project, fulfilling the CPEG/ELEG 499 Engineering Standards requirement. This section draws from the Engineering Standards Report.

Standard	Title / Description	Application in Light Mate
IEEE 802.11 (Wi-Fi)	IEEE Standard for Wireless LAN MAC and PHY	Raspberry Pi 4 built-in 802.11ac used for Lichess API. Defines channel, frequency, and security protocols (WPA2/WPA3).
IEEE 91 / ANSI Y32.2	Graphic Symbols for Electrical Diagrams	All KiCad schematic symbols (resistors, capacitors, ICs, connectors) use IEEE 91/ANSI Y32.2 compliant symbols.
IEEE 315	Graphic Symbols for Electrical Wiring Diagrams	Used with IEEE 91 for all wiring diagrams and block diagrams in this report and schematic package.
IPC-2221B	Generic Standard on Printed Board Design	PCB trace widths follow IPC-2221B charts for 1 oz/ft ² copper. Power traces sized for $\geq 1.62A$. Via sizing meets IPC-2221B minimums.
IPC-A-600	Acceptability of Printed Boards	Fabricated PCBs evaluated against IPC-A-600 Class 2: no measles, no laminate voids, trace adhesion, solder mask coverage.
NXP UM10204	I2C-Bus Specification and User Manual (Rev. 7.0)	I2C implemented at 400 kHz Fast-mode. Pull-up sizing (2.2 k Ω) calculated from UM10204 equations for bus capacitance and rise time.
SPI (Motorola/NXP)	Serial Peripheral Interface Bus Specification	SPI implemented Mode 0 (CPOL=0, CPHA=0) for SK9822 LEDs. Clock polarity, phase, and bit order match datasheet.
UL 217 / IEC 60950	SELV Electrical Safety Standards	All circuitry at SELV levels. No UL listing required at prototype stage but design conforms to SELV principles.
NIST SP 800-series	NIST Information Security Guidelines	API token security practices: Lichess tokens stored in restricted-permission files; not hardcoded in public repositories.
ANSI/IEC 60068	Environmental Testing Standards	Operating temp range (0-40 deg C) per IEC 60068. Enclosure ventilation keeps PCB surface below 60 deg C at full LED load.
ASTM A380	Cleaning and Passivation of Stainless Steel	18-8 SS heat-set inserts and screws (M3, M2.5) sourced per ASTM A380 from McMaster-Carr (97163A149, 94459A783, 92010A116, 92010A016).
ASTM D638 / D790	Tensile/Flexural Properties of Plastics	3D-printed PLA/ABS enclosure mechanical properties referenced against ASTM D638 to confirm strength for mounting loads.
FCC Part 15	Radio Frequency Devices - Unintentional Radiators	Not yet certified (academic prototype). Raspberry Pi 4 carries FCC ID 2ABCB-RPI4B. Full FCC testing required for commercial production.

FIDE Laws of Chess (2023)	World Chess Federation Official Rules	python-chess implements FIDE Laws (2023) including all legal moves, special rules, draw conditions, and checkmate/stalemate detection.
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Table 7: Engineering Standards Applied in Light Mate Design

Adherence to these standards ensures that Light Mate's design is grounded in industry-recognized practices, interoperable with standard tooling and protocols, and ready for future certification if commercialization is pursued.

14. References

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- [11] Lichess.org, Lichess Board API Documentation. Available: <https://lichess.org/api>. Accessed: May 2026.
- [12] python-chess contributors, python-chess: A Chess Library for Python, v1.10.0. Available: <https://python-chess.readthedocs.io>.
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- [14] Raspberry Pi Ltd., Raspberry Pi 4 Model B Datasheet, 2019.
- [15] FIDE, FIDE Laws of Chess 2023, World Chess Federation, 2023.
- [16] McMaster-Carr, Product Specifications: 97163A149, 94459A783, 92010A116, 92010A016. Available: <https://www.mcmaster.com>.
- [17] ASTM International, ASTM A380/A380M-17: Cleaning, Descaling, and Passivation of Stainless Steel, 2017.
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Appendix A: Equipment List

Component	Part Number	Qty	Unit	Total
Hall-Effect Sensor (TO-92)	DRV5033AJQDBZR	200	\$0.25	\$50.00
N52 Neodymium Magnets 20x3mm	Amazon B07CVLXRBW	30	\$0.50	\$14.99
I2C I/O Expander DIP-28	MCP23017-E/SP	7	\$1.99	\$13.90
SK9822 DotStar LED 5050 (pk/10)	COM-14863 SparkFun	120	\$0.45ea	\$54.00
Quad Bus Buffer	SN74HCT125N	2	\$0.76	\$1.52
Raspberry Pi 4 Model B 4GB	SC0194	1	\$55.00	\$55.00
3.5" HDMI Touchscreen	WaveShare	1	\$40.00	\$40.00
5V/5A USB-C Power Supply	Generic	1	\$12.00	\$12.00
2.2kΩ Resistors (pack)	Std 1/4W	50	\$0.02	\$1.00
100nF Decoupling Caps	SMD 0805	50	\$0.02	\$1.00
SparkFun Directional Pad (Qwiic)	COM-14899	1	\$9.50	\$9.50
Heat-set Inserts M3x0.5 SS	McMaster 97163A149	18	\$0.18	\$3.24
Heat-set Inserts M2.5x0.45 SS	McMaster 94459A783	8	\$0.20	\$1.60
Flat Head Screws M3x0.5 (pk25)	McMaster 92010A116	25	\$0.12	\$3.00
Flat Head Screws M2.5x0.45 (pk25)	McMaster 92010A016	25	\$0.10	\$2.50
4x4 Prototype PCB (JLCPCB)	Custom KiCad	5	\$9.64	\$48.18
8x8 Final PCB (JLCPCB)	Custom KiCad	5	\$15.36	\$76.78
PLA/ABS Filament (enclosure)	Generic 1.75mm	1 kg	\$22.00	\$22.00
Misc. (wire, solder, flux)	--	--	--	Provided by University

Table A1: Full Equipment and Component List

Appendix B: Budget Summary

The project was funded by a \$250 team budget. The following table summarizes actual expenditures.

Category	Amount	Notes
Components (Fall 2025)	\$94.95	Hall sensors, magnets, MCP23017s, LEDs, level shifter
PCB - 4x4 Prototype	\$48.18	JLCPCB, 5-board panel, 160x130mm, 2-layer
PCB - 8x8 Final Board	\$76.78	JLCPCB, 5-board panel, 280x250mm, 2-layer
Additional Spring 2026	~\$76.00	Replacement LEDs (100), Hall sensors (100), Qwiic D-pad, hardware
Enclosure (3D printing)	~\$22.00	PLA/ABS filament, McMaster hardware fasteners
Subtotal (core)	\$219.91	Fall 2025 components + both PCBs
Spring additions	~\$98.00	Additional components + enclosure materials
TOTAL SPENT (est.)	~\$317.91	Slightly over \$250; Spring extras funded by University, some partially self-funded by team
Original Budget	\$250.00	University-allocated team budget
Over/Under	+\$67.91	Over budget

Table B1: Project Budget Summary

-- End of Report --

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